

Functional Requirements Document (FRD)

1. Introduction

This Functional Requirements Document (FRD) describes the detailed functional behavior of the Online Food Delivery Platform. It translates business requirements defined in the BRD into specific functional specifications that describe how the system will operate from a user and system interaction perspective.

2. Scope Reference

This document covers functional requirements related to user registration, restaurant browsing, order placement, payment processing, order tracking, restaurant order management, and administrative controls.

3. User Roles

Role	Description
Customer	End user who places food orders
Restaurant User	Receives and manages food orders
Delivery Partner	Delivers food orders
Admin	Manages platform operations

4. Functional Overview

The system shall enable customers to browse restaurants, select menu items, place orders, make payments, and track order status. Restaurants shall receive orders digitally and update order statuses. Administrators shall manage users, restaurants, and reports.

5. Functional Requirements

5.1 Customer Registration and Login

FR ID	Functional Requirement
FR-01	The system shall allow customers to register using mobile number or email
FR-02	The system shall verify user identity through OTP
FR-03	The system shall allow registered users to log in
FR-04	The system shall provide password reset functionality

5.2 Restaurant Browsing

FR ID	Functional Requirement
FR-05	The system shall display restaurants based on customer location

FR-06	The system shall allow filtering by cuisine and rating
FR-07	The system shall display restaurant details including menu and delivery time

5.3 Menu Selection and Cart Management

FR ID	Functional Requirement
FR-08	The system shall display menu items with prices
FR-09	The system shall allow users to add or remove items from the cart
FR-10	The system shall calculate total order value in real time

5.4 Order Placement

FR ID	Functional Requirement
FR-11	The system shall allow users to place orders from the cart
FR-12	The system shall capture delivery address details
FR-13	The system shall generate a unique order ID

5.5 Payment Processing

FR ID	Functional Requirement
FR-14	The system shall provide multiple payment options
FR-15	The system shall process online payments securely
FR-16	The system shall display payment success or failure message

5.6 Order Tracking

FR ID	Functional Requirement
FR-17	The system shall display real-time order status
FR-18	The system shall notify users of order status changes
FR-19	The system shall display estimated delivery time

5.7 Restaurant Order Management

FR ID	Functional Requirement
FR-20	The system shall notify restaurants of new orders

FR-21	Restaurants shall accept or reject orders
FR-22	Restaurants shall update order preparation status

5.8 Admin Functions

FR ID	Functional Requirement
FR-23	Admin shall onboard and manage restaurants
FR-24	Admin shall manage user accounts
FR-25	Admin shall view platform reports

6. Business Rules

Rule ID	Business Rule Description
BR-01	Orders cannot be canceled once preparation has started
BR-02	Cash on delivery is allowed only below a defined order value
BR-03	Failed payments shall cancel the order after three attempts

7. Error Handling

Scenario	System Response
Payment failure	Display error and allow retry
Restaurant rejects order	Notify customer and initiate refund
Network issue	Display retry option

8. Assumptions and Dependencies

Type	Description
Assumption	Users have stable internet connectivity
Dependency	Third-party payment gateway integration
Dependency	SMS/Email notification services

9. Traceability Matrix

BRD ID	FRD ID
BR-01	FR-01 to FR-04
BR-02	FR-05 to FR-07
BR-03	FR-08 to FR-13
BR-04	FR-14 to FR-16
BR-05	FR-17 to FR-22